Strange Attractors in Gradient Descent: Data Structure and Loss Geometry Control Fractal Dimension

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We measure the correlation dimension D_2 of neural network training trajectories in function space. A convolutional network trained on CIFAR-10 under full-batch gradient descent produces a strange attractor with $D_2 = 3.67 \pm 0.08$ (conservative lower bound; convergence analysis yields $D_2 \approx 5$). Cross-architecture controls show that structured data is necessary for multi-dimensional chaos—either architecture on synthetic data gives $D_2 \approx 1$—while architecture modulates the learning-rate threshold at which the transition occurs. A label-noise sweep reveals two distinct mechanisms: at moderate learning rates, degrading label structure reduces D_2 from 3.6 to 1.7 as data-created oscillatory modes are destroyed; near the Edge of Stability, loss-surface roughness from random labels sustains high-dimensional dynamics independently. Peak perturbation sensitivity and peak geometric complexity occur at different learning rates across all architectures, with the richest structure emerging where chaotic and convergent dynamics compete. The attractor survives stochastic gradient descent, retaining >82% of its fractal dimension at 20× gradient noise.

INTRODUCTION

Gradient descent is a deterministic dynamical system. Each update maps the network's function to a new function, producing a trajectory through function space whose geometry can be measured.

Recent work has established that this geometry is nontrivial. Cohen *et al.* [1] discovered the Edge of Stability (EoS): the top Hessian eigenvalue rises to $2/\eta$ and oscillates rather than diverging. Damian *et al.* [2] characterized the self-stabilization mechanism. Kalra *et al.* [3] and Ghosh *et al.* [4] observed period-doubling near the EoS, and Morales *et al.* [5] computed Lyapunov exponents during training, establishing that the dynamics can be chaotic. Storm *et al.* [6] showed that finite-time Lyapunov exponents through network depth form coherent structures in input space.

These results establish that training dynamics are rich, but leave open a geometric question: what *shape* does the chaos have? A chaotic trajectory might be essentially one-dimensional—erratic motion along a single convergence path—or it might fill a multi-dimensional fractal volume. We measure this distinction using the correlation dimension D_2 [7] and find that training trajectories settle onto strange attractors whose fractal dimension is shaped by two mechanisms: structured data creates oscillatory modes at moderate learning rates, and loss-surface roughness sustains high-dimensional chaos near the Edge of Stability. By *data complexity* we mean the degree to which the input-label mapping requires the network to maintain multiple competing internal representations—operationally, the mutual information between inputs and labels modulated by the geometric structure of the input distribution.

EXPERIMENTAL FRAMEWORK

Architectures and data.—We study five conditions under full-batch gradient descent with MSE loss, no momentum, and no weight decay (Table I). Full-batch training isolates the deterministic dynamics. Learning rates are expressed as fractions of $2/\lambda_{\max}$, where $\lambda_{\max}$ is the top Hessian eigenvalue after 1,000 warmup steps at $\eta = 0.01$. All networks are trained for 5,000 steps.

The CNN has two convolutional layers (3→16→32 channels, 3×3 kernels, max pooling, ReLU, two fully connected layers; 268,650 parameters) trained on a 2,000-image CIFAR-10 subset. The MLPs use 2-hidden-layer tanh networks of width 50 (156K parameters) or 85 (269K parameters) on either CIFAR-10 (flattened) or synthetic data (220 dimensions, 10 classes). The CNN-on-synthetic condition embeds synthetic data into 3×32×32 tensors to isolate architectural effects.

Lyapunov exponents.—We measure chaos in function space to avoid overparameterization artifacts. Two copies of the network train from identical initialization, one perturbed by $\varepsilon = 10^{-5}$ along a random unit-norm direction. The function-space divergence $\delta(t) = \|f(\mathbf{x}) - \tilde{f}(\mathbf{x})\|$ is recorded on 100 held-out inputs; λ is the slope of $\ln \delta(t)$ versus t over $[0.2T, 0.8T]$, with t measured in training steps so that λ has units of inverse steps. Sensitivity analysis confirms that $\varepsilon = 10^{-8}$ produces uniform artifactual positive exponents; $\varepsilon \geq 10^{-6}$ is required for converged measurements. The choices $\varepsilon = 10^{-5}$ and 100 held-out inputs are within a plateau verified in the supplement [8]. This procedure measures a finite-scale, finite-time divergence rate; the ε-dependence documented in [8] implies systematic uncertainty of approximately a factor of 3 beyond the seed-to-seed variability reported here. Results are averaged over 10 seeds (CNN/CIFAR-10, MLP/CIFAR-10, CNN/synthetic) or 20 seeds (MLP/synthetic).

Correlation dimension.—D_2 is computed via the Grassberger-Procaccia algorithm [7] on function-space trajectories (outputs recorded every 10 steps, first 20% discarded as transient, yielding ∼400 points). Pipeline validation recovers $D_2 = 2.18$ for a quasiperiodic 2-torus (expected 2.0) and $D_2 = 1.73$ for the Lorenz attractor (expected 2.05). Extended calibration [8] reveals a systematic 15–35% underestimate for fractal attractors; a $D_2(N)$ convergence analysis confirms that reported values are conservative lower bounds that have not yet saturated at $N \approx 400$ [8]. The first 20% of each trajectory is discarded as transient; at high learning rates where sharpness may not have fully equilibrated within 5,000 steps, the measurements characterize the quasi-stationary dynamics of the approach to equilibrium.

Persistent homology.—Computed via RIPSER [9] on PCA-reduced trajectories to distinguish smooth tori (few persistent loops, large gap ratio) from strange attractors (many features at all scales, gap ratio ≈1).

RESULTS

The chaos window.—The Lyapunov divergence rate is non-monotonic with learning rate [Fig. 1(a)]. In the CNN on CIFAR-10, λ peaks at 15% of the EoS threshold and crosses zero near 45%. Above this, basin convergence dominates: trajectories from different perturbations collapse to the same learned function. This bounded chaos window appears in every architecture tested; its width scales with architecture complexity.

Training trajectories are strange attractors.—Within the chaos window, the CNN/CIFAR-10 trajectory acquires fractal structure [Fig. 1(b)]. D_2 crosses 1.0 at 10% of EoS, rises through 2.63 ± 0.62 at 20%, peaks at 3.67 ± 0.08 at 30%, and remains above 2.0 to 80% of EoS. The first principal component captures only ∼52% of trajectory variance—half the dynamics are off the primary convergence axis. A convergence analysis extending trajectories to $N = 3200$ points [8] shows that D_2 has not saturated at the production length $N \approx 400$: both MLPs stabilize near $D_2 \approx 5.1$, confirming that the $N \approx 400$ values reported here are conservative lower bounds. Accounting for the 15–35% systematic underestimate from our calibration [8], the true attractor dimensions for CIFAR-10 conditions likely lie in the range $D_2 \approx 4$–7.

Persistent homology confirms fractal topology across architectures [8]. For the CNN at 30% of EoS, 386 H_1 features appear with gap ratio 1.2 (no dominant loops); at 5% of EoS, zero H_1 features: a simple convergence path. Both MLPs on CIFAR-10 show the same transition: zero H_1 features below 30% of EoS, rising to 389 (H_1) and 337 (H_2) features at 90% for the 156K MLP, comparable to the CNN at its peak. The gap ratio remains near 1.0 in every condition—the topological signature of a strange attractor, not a smooth torus.

Structured data is necessary; architecture modulates the threshold.—The cross-architecture controls (Table I, Fig. 4) reveal a clean separation. A CNN on structureless synthetic data produces $D_2 < 1.0$ at all learning rates—architecture alone is insufficient. Both MLPs on CIFAR-10 reach $D_2 > 4$ at 90% of EoS (converging to $D_2 \approx 5.1$ at longer trajectory lengths); the small difference between the 156K and 269K networks at $N \approx 400$ (4.61 ± 0.10 vs 4.37 ± 0.05) has not converged and should not be over-interpreted. Structured data is both necessary (CNN + synthetic → $D_2 \approx 1$) and sufficient (MLP + CIFAR-10 → $D_2 > 4$) for multi-dimensional chaos.

Architecture modulates the *threshold*: at 30% of EoS the CNN has already reached $D_2 = 3.67$ while both MLPs remain at $D_2 \approx 1$. The MLPs require 70–90% of EoS to achieve comparable fractal dimension. The convolutional hierarchy facilitates mode coupling at lower learning rates.

Two mechanisms control attractor dimension.—A label-noise sweep from $p = 0$ (clean CIFAR-10) to $p = 1$ (random labels) at each architecture's peak-D_2 learning rate (Fig. 2) reveals that fractal dimension is shaped by two distinct mechanisms depending on the dynamical regime. At moderate learning rates (CNN at 30% EoS), structured labels create the oscillatory modes that the attractor fills: D_2 falls smoothly from 3.64 ± 0.11 to 1.68 ± 0.54 as label structure is degraded. Near the Edge of Stability (MLP at 90% EoS), a different mechanism operates: random labels roughen the loss surface, creating competing minima that sustain higher-dimensional dynamics even without label structure, and D_2 rises modestly from 4.36 ± 0.05 to 4.77 ± 0.08. In both regimes, changing label content continuously changes the attractor, but the sign of the response reverses—a signature of two competing sources of dynamical complexity.

The attractor survives stochastic gradient descent.—The preceding measurements use full-batch gradient descent; we test robustness by sweeping batch size B from 2000 (full-batch) down to 100 for both the CNN at 30% of EoS and the MLP (269K) at 90% of EoS (Fig. 3). Both the original and perturbed model copies see identical mini-batches at each step (paired-noise protocol), so the Lyapunov measurement isolates deterministic sensitivity from stochastic forcing. The attractor dimension degrades gracefully in both architectures: the CNN retains 87% of its full-batch D_2 at $B = 100$ (3.20 ± 0.05 versus 3.67 ± 0.10); the MLP retains 82% (3.59 ± 0.08 versus 4.37 ± 0.01). Both remain well above $D_2 = 3$ at all batch sizes tested. Meanwhile, λ *increases* under SGD—by 10× for the CNN (peaking at $B = 500$) and by the same factor for the MLP (peaking at $B = 200$): stochastic forcing amplifies perturbation sensitivity along the existing attractor modes without creating new dynamical dimensions.

Peak chaos and peak complexity occur at different

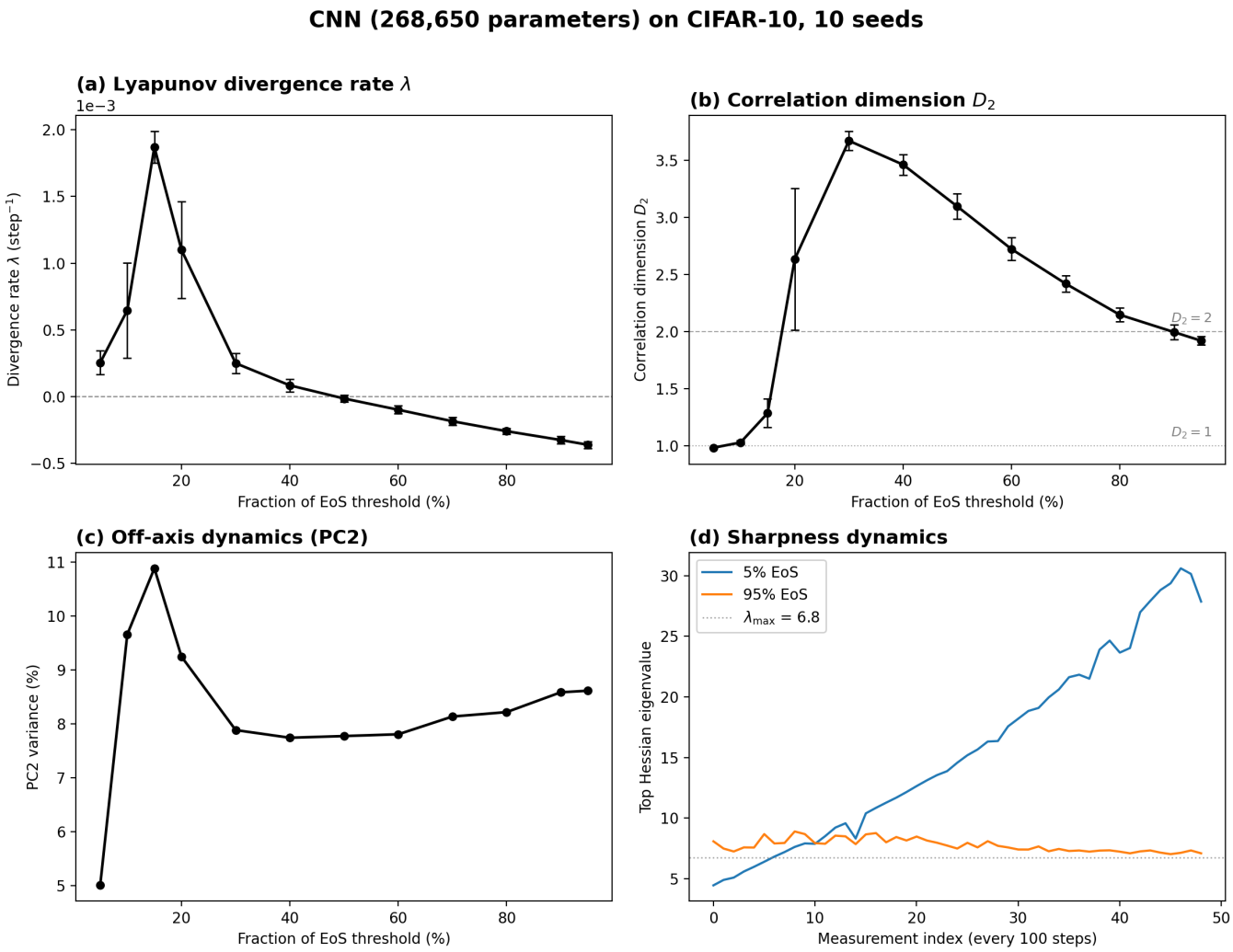

FIG. 1. CNN (268,650 parameters) on CIFAR-10, 10 seeds. (a) Finite-scale Lyapunov divergence rate λ versus fraction of the EoS threshold $2/\lambda_{\max}$. Positive λ peaks at 15% and crosses zero near 45%. (b) Correlation dimension D_2 versus EoS fraction. D_2 crosses 1 at 10% and peaks at 3.67 ± 0.08 at 30% ($N \approx 400$; conservative lower bound). (c) Second principal component variance, showing off-axis dynamics. (d) Top Hessian eigenvalue: monotonic progressive sharpening at 5% EoS (blue) versus bounded oscillation at 95% EoS (orange). Error bars: standard deviation across seeds.

learning rates.—For the CNN on CIFAR-10, peak λ occurs at 15% of EoS while peak D_2 appears at 30%. The same dissociation appears across all architectures: the MLP (156K) peaks in λ at 40% versus D_2 at 90% of EoS; the MLP (269K) at 50% versus 90%. In every case the attractor is geometrically most complex not where perturbation sensitivity is greatest, but in the transition zone where chaotic dynamics and basin convergence compete. This phenomenology is reminiscent of transitions in near-integrable systems [10], where ordered and chaotic regions coexist, though we note that gradient descent is not Hamiltonian and the analogy is phenomenological rather than rigorous.

TABLE I. Experimental conditions with D_2 at each condition's peak. The $N{\approx}400$ column reports production measurements; the $N{=}3200$ column reports converged values from the trajectory-length analysis [8].

	Params	Data	D_2 ($N{\approx}400$)	D_2 ($N{=}3200$)	EoS%	Seeds
MLP/synth.	14K	synth.	0.9	—	all	20
CNN/synth.	269K	synth.	0.98(3)	—	90	10
MLP/CIFAR	156K	C-10	4.61(10)	∼5.1	90	10
MLP/CIFAR	269K	C-10	4.37(5)	∼5.1	90	10
CNN/CIFAR	269K	C-10	3.67(8)	∼5*	30	10

*CNN convergence from single seed; noisier than MLP conditions.

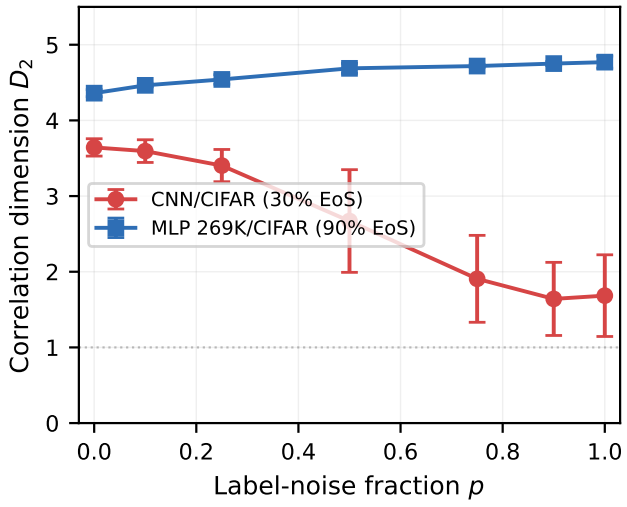

FIG. 2. D_2 versus label-noise fraction p at each architecture's peak learning rate. CNN at 30% EoS (red circles): D_2 falls as label structure is degraded. MLP 269K at 90% EoS (blue squares): D_2 rises modestly as random labels roughen the loss landscape. Three seeds per condition; error bars (standard deviation) may underestimate true variability at this sample size.

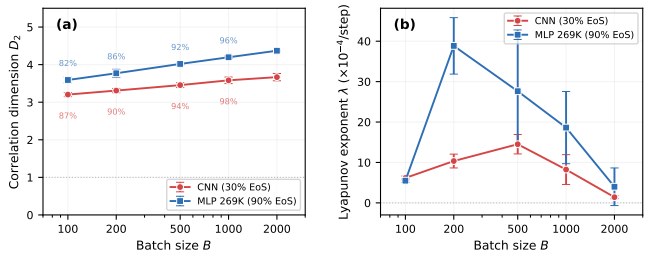

FIG. 3. Batch-size sweep for CNN/CIFAR-10 at 30% EoS (red circles) and MLP 269K at 90% EoS (blue squares). (a) Correlation dimension D_2 versus batch size B. Percentages indicate retention relative to full-batch ($B = 2000$). Both architectures remain above $D_2 = 3$ at all batch sizes. (b) Lyapunov divergence rate λ versus B. SGD noise amplifies λ by up to 10× in both architectures. Three seeds per condition; error bars (standard deviation) may underestimate true variability at this sample size.

DISCUSSION

These results provide a geometric characterization of gradient descent dynamics. Training trajectories settle onto strange attractors whose fractal dimension is shaped by two mechanisms: at moderate learning rates, structured data creates oscillatory modes that fill a multi-dimensional volume; near the Edge of Stability, loss-surface roughness—whether from structured or random labels—sustains high-dimensional chaos through competing minima. Architecture modulates the learning-rate threshold at which the transition occurs but does not determine the attractor's destination.

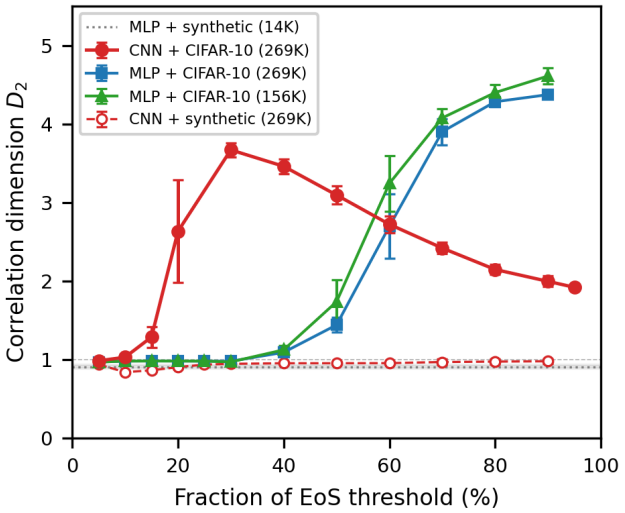

FIG. 4. D_2 versus fraction of the EoS threshold for all five conditions. CNN on CIFAR-10 (red circles) peaks at 30% EoS; MLPs on CIFAR-10 (blue squares, green triangles) peak at 90% EoS. Synthetic conditions (open markers) remain at $D_2 \approx 1$ throughout. All CIFAR and CNN/synthetic conditions: 10 seeds; MLP/synthetic: 20 seeds. Error bars: standard deviation across seeds.

The label-noise sweep (Fig. 2) provides the clearest evidence for these two mechanisms. The CNN at 30% of EoS shows the data-structure mechanism in isolation: degrading labels smoothly destroys the oscillatory modes and collapses D_2 toward the synthetic baseline. The MLP at 90% of EoS reveals the loss-geometry mechanism: even random labels sustain high-dimensional dynamics because the rougher loss surface provides its own competing modes. The sign reversal of D_2's response to label noise across regimes is a distinctive signature that any theoretical account of these dynamics must explain.

The findings are suggestive of classical routes to chaos such as Newhouse-Ruelle-Takens [11, 12]: complex data or rough landscapes create independent axes of oscillation, and when enough modes couple through the training dynamics, the flow produces a strange attractor. The bounded chaos window and the universality of the λ–D_2 dissociation across architectures—each showing peak geometric complexity offset from peak sensitivity—are reminiscent of the phenomenology of transitions in near-integrable systems [10], where ordered and chaotic regions coexist. However, gradient descent is not Hamiltonian, and these analogies are phenomenological rather than rigorous identifications. The CNN's hierarchical processing creates efficient coupling pathways, lowering the learning rate at which the transition occurs. The MLP accesses the same modes—provided by the data—but requires stronger driving to couple them. On structureless data, no architecture produces multi-dimensional

dynamics because the modes do not exist.

These dynamical findings complement Pittorino *et al.* [13], who showed that the loss landscape has toroidal topology after symmetry removal—a static geometric property of the parameter space on which these dynamical transitions unfold.

A methodological note: our λ values are finite-scale, finite-time divergence rates rather than true Lyapunov exponents (which require $\varepsilon \to 0$, $T \to \infty$). Measurements at perturbation scale $\varepsilon = 10^{-8}$ are dominated by numerical noise [8]; reliable measurements require $\varepsilon \geq 10^{-6}$. Prior work at smaller ε [5] should be interpreted with caution. The function-space protocol avoids false positives from gauge symmetries in overparameterized networks.

The batch-size sweep (Fig. 3) establishes that the strange attractor is not an artifact of full-batch training. Reducing B from 2000 to 100 introduces substantial stochastic forcing, yet both architectures retain >82% of their fractal dimension. The order-of-magnitude increase in λ under SGD while D_2 changes modestly reveals a separation of scales: gradient noise amplifies sensitivity along the attractor's existing modes without generating new dynamical dimensions. This is consistent with the stochastic forcing acting as a perturbation to the deterministic skeleton rather than restructuring the dynamics.

Several questions remain open. Does D_2 continue to grow with larger architectures and datasets? The learning rates that produce strange attractors overlap with those commonly used in practice [1]. Whether this connection is causal—whether exploring a higher-dimensional attractor during training produces better solutions—is the central question these measurements now enable.

AI language models (Claude, Anthropic) were used as research tools throughout this work, including assistance with experimental design, code development, data analysis, and manuscript preparation. All research directions, experimental choices, and scientific conclusions are the sole responsibility of the author. Source code, raw data, and analysis scripts are available at `https://github.com/gnomevan/attractor`.

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[1] J. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar, Gradient descent on neural networks typically occurs at the edge of stability, Proceedings of the International Conference on Learning Representations (2021).

[2] A. Damian, E. Nichani, and J. D. Lee, Self-stabilization: The implicit bias of gradient descent at the edge of stability, Proceedings of the International Conference on Learning Representations (2023).

[3] D. S. Kalra, T. He, and M. Barkeshli, Universal sharpness dynamics in neural network training: Fixed point analysis, edge of stability, and route to chaos, Proceedings of the International Conference on Learning Representations (2025), arXiv:2311.02076.

[4] A. Ghosh, S. M. Kwon, R. Wang, S. Ravishankar, and Q. Qu, Learning dynamics of deep linear networks beyond the edge of stability, Proceedings of the International Conference on Learning Representations (2025), arXiv:2502.20531.

[5] A. Morales, Y. Rosas-Guevara, and J. C. Toledo-Roy, Lyapunov exponents during training of neural networks, Frontiers in Physics **12**, 10.3389/fphy.2024.1436464 (2024).

[6] L. Storm, H. Linander, J. Bec, K. Gustavsson, and B. Mehlig, Finite-time Lyapunov exponents of deep neural networks, Phys. Rev. Lett. **132**, 057301 (2024).

[7] P. Grassberger and I. Procaccia, Characterization of strange attractors, Phys. Rev. Lett. **50**, 346 (1983).

[8] See Supplemental Material at [URL will be inserted by publisher] for perturbation sensitivity analysis, D_2 pipeline calibration on known attractors (Hénon, Lorenz, Rössler, Mackey-Glass, smooth tori), extended D_2 values at all learning rate fractions, persistent homology details for all architectures, D_2 convergence with trajectory length, convergence of D_2 and λ with measurement parameters, depth scaling, label-noise sweep data, λ–D_2 dissociation analysis, and batch-size sweep details.

[9] C. Tralie, N. Saul, and R. Bar-On, Ripser.py: A lean persistent homology library for Python, J. Open Source Softw. **3**, 925 (2018).

[10] A. N. Kolmogorov, On the conservation of conditionally periodic motions under small perturbation of the Hamiltonian (1954), see also V. I. Arnold, Russ. Math. Surv. **18**, 85 (1963); J. Moser, Nachr. Akad. Wiss. Göttingen II, 1 (1962).

[11] D. Ruelle and F. Takens, On the nature of turbulence, Commun. Math. Phys. **20**, 167 (1971).

[12] S. Newhouse, D. Ruelle, and F. Takens, Occurrence of strange Axiom A attractors near quasi periodic flows on T^m, $m \geq 3$, Commun. Math. Phys. **64**, 35 (1978).

[13] F. Pittorino, A. Ferraro, G. Perugini, C. Feinauer, C. Baldassi, and R. Zecchina, Deep networks on toroids: Removing symmetries reveals the structure of flat regions in the landscape geometry, in *Proceedings of the 39th International Conference on Machine Learning*, PMLR, Vol. 162 (2022) pp. 17759–17781, arXiv:2202.03038.